

From a LEEM focal series to the object: a tutorial

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Abstract

A low-energy electron microscope records pictures: brightness at each pixel, at several focus settings. Those pictures are not the sample. The sample is encoded in a complex wave that left the surface; the camera keeps only the brightness of that wave after the lenses have blurred, filtered, and mixed its spatial frequencies. This note explains, for a reader who has not met Fourier optics, numerical inverse methods, or the Lean proof assistant, how to go backwards: from a through-focal stack of intensities to the exit wave $\psi_0 = \sigma e^{i\phi}$, and from ϕ to topographic height in reflection. Yu, Lau and Altman [11] wrote the forward map (object to image) and matched contrast; they did not invert. The pipeline recommended here is two-stage. Stage 1 is a regularized, frequency-by-frequency inversion of the through-focal stack, using the same kernel as Yu *et al.* restricted to the contrast-transfer slice. It is the unique published style of 2D LEEM exit-wave reconstruction (Duden, Thust, Kumpf and Tautz [12]) and it is the fastest method that still fills the holes a single focus leaves in the contrast-transfer function. Stage 2 is an optional nonlinear refinement for strong-phase objects (graphene ripples, atomic steps); it is skipped when the stage-1 prediction already matches the stack to within the noise ($\max|\phi| \lesssim 0.3$). A one-dimensional sinusoidal ripple, the example Yu *et al.* used to exhibit contrast-transfer failure, reduces stage 2 to fitting a short ladder of Bessel amplitudes. Lean 4 is used as a certificate that the discrete linear stage is well-posed under that kernel—unique answer, exact error identity, proved physical blind spots—not as a replacement for the experiment.

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1 Who this is for

This note assumes you know what a microscope image is and why one might record several of them at different focus. It does not assume Fourier optics, numerical analysis, or any experience with proof assistants.

Three ideas are introduced in order. First the experiment: what a LEEM records, and what we want back (Sections 2 to 4). Then the optical model that sits between those two things, in one page of Fourier optics (Sections 5 and 6). Then Lean: what a machine-checked proof is, why it is useful here, and exactly which claims about the numerical method have certificates (Sections 7 and 8). Only after that do we compare reconstruction methods and write down the recipe (Sections 9 to 13).

Equations are translations of the prose, not a substitute for it. Formal theorem statements live in Appendix A; they are optional. The companion checklist for the machine-checked statements is docs/LINEAR_INVERSE.md in this repository.

2 What the experiment records

Low-energy electron microscopy (LEEM) is a surface microscope. A slow electron beam (typically a few electronvolts at the sample) reflects from a crystalline surface and is then accelerated to column energy, about 15 kV, and imaged by electron lenses [1, 2, 3, 4]. Aberration-corrected instruments of the kind Yu, Lau and Altman [11] model are described by Tromp and coworkers [5, 6].

The detector is a camera. At each pixel it records a real, nonnegative number: how many electrons arrived, or equivalently the *intensity* $I(\mathbf{r})$. It does not record the electron wave. The wave has an amplitude and a phase; the camera keeps the squared modulus after the lenses have acted, and throws the phase away. That is the same limitation as an ordinary light camera, and it is the reason reconstruction is a problem at all.

A *through-focal series*, or focal stack, is the same field of view recorded at several focus settings Δz_k , $k = 1, \dots, K$. Typical numbers in this note are $K = 11$ –21 planes and $N = 512^2$ pixels. Changing focus changes how the lenses delay off-axis rays relative to the axis, so the same object produces a different interference pattern in each plane. Those differences are the extra information that makes inversion possible.

Yu, Lau and Altman [11], following Pang *et al.* [7] and Schramm *et al.* [8], write a precise forward map from a trial object to such a stack and compare the predicted contrast with experiment. That is *forward matching*. This note is about the inverse: given the stack, recover the object.

3 What we want back

Just above the surface the reflected electrons form a complex *exit wave*

$$\psi_0(\mathbf{r}) = \sigma(\mathbf{r}) e^{i\phi(\mathbf{r})}. \quad (1)$$

Here σ is a real amplitude and ϕ is a real phase in *radians*. For a topographic height A seen in reflection, the kinematic phase is

$$\phi = \frac{4\pi A}{\lambda_0}, \quad (2)$$

where λ_0 is the electron wavelength *at the sample*. At $E_0 = 10$ eV, $\lambda_0 \simeq 0.39$ nm, so a height of 0.1 nm already gives $\phi \sim 3$: not a small perturbation. Graphene ripples and atomic steps live in that regime [10].

Two wavelengths appear in this subject, and mixing them is the most common modelling error.

- λ_0 is the wavelength at the sample (a few electronvolts). It sets the object phase (2).
- λ is the wavelength *after acceleration* to column energy E (about 15 kV). It is a few picometres. It sets the lens optics: defocus, spherical aberration, and the meaning of a spatial frequency $\mathbf{q} = \boldsymbol{\alpha}/\lambda$ [9, 8].

The reconstruction kernel below uses λ . The conversion of reconstructed phase to height uses λ_0 .

The Fourier transform of ψ_0 is written Ψ . A spatial frequency $\mathbf{q}$ is a sinusoidal ripple across the field of view: large $|\mathbf{q}|$ is fine detail.

4 Why one picture is not enough

Four independent losses of information sit between ψ_0 and a single image $I(\mathbf{r}, \Delta z)$.

The camera drops phase. Intensity is $|\text{wave in the image plane}|^2$. Any global factor $e^{i\theta}$ on ψ_0 cancels and never appears. The reconstruction can at best recover ψ_0 up to one overall phase, which we fix by a convention (the “vacuum gauge” of Section 11).

The lenses filter. Each spatial frequency is multiplied by a complex factor that depends on defocus and on residual aberrations. That factor—introduced properly in Section 5—can be zero at some $\mathbf{q}$. Dividing the data by a zero is meaningless: those frequencies are absent from that image.

The contrast aperture cuts. A physical aperture in the column passes only $|\mathbf{q}| \leq q_{\text{ap}}$. Finer detail is discarded before the camera. No later algorithm can invent it. Those modes reconstruct as zero, and that is a prediction, not a bug.

One complex measurement of two unknowns. A weak perturbation of a strong reflected beam produces, at each frequency $\mathbf{q}$, a pair of unknown complex amplitudes $(X(\mathbf{q}), X(-\mathbf{q}))$. One image supplies one complex equation for that pair, so a whole line of objects give the same picture. A second, independent focus setting supplies a second equation.

The practical conclusion: record a stack with $K \geq 2$ defoci whose filter zeros do not sit on top of each other, and do not ask the reconstruction for detail beyond q_{ap} or for an absolute global phase.

5 Fourier optics in one page

Fourier optics is the statement that an image is interference. Goodman’s textbook [13] is the standard reference in light optics; the LEEM versions are Pang *et al.* [7], Schramm *et al.* [8], Yu *et al.* [10, 11], and the χ -in-waves convention of Tromp and Schramm [9].

Decompose the exit wave into spatial frequencies $\Psi(\mathbf{q})$. In the image plane, every pair of frequencies $(\mathbf{q}, \mathbf{q}')$ beats against each other and produces a fringe at $\mathbf{q} - \mathbf{q}'$. The strength of that fringe is a complex weight $R_{\text{FO}}(\mathbf{q}, \mathbf{q}', \Delta z)$, the *Fourier-optics kernel*. The recorded intensity is the sum of all such fringes:

$$I(\mathbf{r}, \Delta z) = \iint \Psi(\mathbf{q}) \Psi^*(\mathbf{q}') R_{\text{FO}}(\mathbf{q}, \mathbf{q}', \Delta z) e^{2\pi i(\mathbf{q}-\mathbf{q}') \cdot \mathbf{r}} d^2 q d^2 q'. \quad (3)$$

The kernel is supported on the aperture $|\mathbf{q}|, |\mathbf{q}'| \leq q_{\text{ap}}$. In the notation of Yu *et al.* [11] it factors as $R_{\text{FO}} = R_0 E_S E_{C,\text{tot}}$: a perfectly coherent piece R_0 times a spatial (source-size) envelope E_S times a chromatic (energy-spread) envelope $E_{C,\text{tot}}$. Those envelopes are closed-form Gaussians; they are proved in the companion note [27] and encoded in this repository.

Two pictures of the same optics are in circulation.

Contrast-transfer function (CTF). Pretend each frequency travels alone: multiply $\Psi(\mathbf{q})$ by a filter $H(\mathbf{q}, \Delta z)$, inverse-transform, and take the modulus squared. This is accurate when the illumination is perfectly coherent, and it remains a good linearization when the object is a *weak* perturbation of a flat mirror (small ϕ). Schramm *et al.* [8] develop that description for aberration-corrected LEEM and PEEM.

Fourier optics (FO). Keep the pairs. Partial coherence—a finite source and a finite energy spread—mixes $\mathbf{q}$ with $\mathbf{q}'$, so the weight $R_{\text{FO}}(\mathbf{q}, \mathbf{q}')$ does not factor as $H(\mathbf{q})\overline{H(\mathbf{q}')}$. Yu *et al.* [10] showed that this mixing is why a CTF calculation misses the contrast of rippled graphene. The FO kernel agrees with the CTF kernel on the slice $\mathbf{q}' = \mathbf{0}$:

$$R_{\text{FO}}(\mathbf{q}, \mathbf{0}, \Delta z) = R_{\text{CTF}}(\mathbf{q}, \Delta z). \quad (4)$$

The other axis $\mathbf{q} = \mathbf{0}$ is the conjugate branch $R_{\text{FO}}(\mathbf{0}, -\mathbf{q})$, not the CTF. Stage 1 below inverts the shared $\mathbf{q}' = \mathbf{0}$ slice. Stage 2 uses the full pair kernel, which is what Yu *et al.* actually compared with experiment.

Lens phase, in waves. The lenses delay a ray by an optical path χ , measured in *waves* (path divided by λ), not in radians. The transfer factor is $W = e^{2\pi i \chi}$ [9]. Defocus enters as $+\frac{1}{2}\Delta z \lambda q^2$ in χ . That sign is a convention of this library: positive Δz adds path for off-axis rays. Some published simulators use the opposite sign; mixing the two swaps overfocus and underfocus without changing any plot label. Do not convert χ to radians and then insert another 2π .

6 What “inverting” means

Given $I(\mathbf{r}, \Delta z_1), \dots, I(\mathbf{r}, \Delta z_K)$, find an exit wave ψ_0 whose predicted stack, through (3), matches the measurements as well as the noise allows.

That is not what Yu, Locatelli and Altman [10] or Yu, Lau and Altman [11] did. They chose a trial object (a step, a sinusoid), ran (3) forwards, and overlaid predicted versus measured contrast against defocus. The only demonstrated 2D experimental LEEM exit-wave reconstruction is Duden, Thust, Kumpf and Tautz [12]: a linear through-focal restoration of a molecular monolayer, in the tradition of Schiske’s focus-series reconstruction in electron optics [14, 15]. Duden *et al.* assume a strong specular beam and a linear (CTF-style) model.

This note is the inverse of the Yu 2019 forward model. Stage 1 is Duden’s algorithm with the FO/CTF envelopes of this library in place of a coherent probe. Stage 2 is the refinement required when the object is too strong for that linearization—the regime that made FO necessary in the first place.

7 What Lean is, and why it is here

A *proof assistant* is a computer program that checks a mathematical argument the way a compiler checks types: every inference must follow from stated definitions and previously accepted lemmas, or the file is rejected. Lean 4 is such a system [23]. Its companion library Mathlib is a large, continuously checked corpus of algebra, analysis, and special functions [24]. This repository uses Lean 4 with Mathlib to encode the Yu 2019 kernel and the discrete reconstruction below. Surveys of why machine-checked proofs matter in scientific computing are given by Harrison [25] and by Avigad and Harrison [26].

That is a different activity from writing a numerical script. A script evaluates numbers: it can plot a through-focal series, invert a matrix, or report a residual. It does not, by itself, certify that the kernel on the screen is the kernel in the paper, that the regularized inverse is the unique

minimizer of the energy we wrote down, or that an error bound is an identity rather than a lucky observation on one dataset.

The LEEM inverse is exactly the setting in which that distinction bites. Papers treat uniqueness, aperture support, and “one image is enough” as routine, and those are the places the conventions of [Sections 3](#) and [5](#) go wrong: λ versus λ_0 , χ in waves, CTF versus FO, and the sign of defocus. A Python reconstruction can run, produce a plausible-looking height map, and still have implemented the wrong kernel. Lean does not replace the experiment or the discrete Fourier transform. It certifies the *algebra of the method*, pinned to the same R_{FO} already proved in this repository [\[27\]](#).

What, then, does it mean to say that the “numerical proof is good”? It does not mean that a reconstruction looks plausible. It means three things a script can illustrate but cannot settle.

1. The regularized estimator is *unique*: for every frequency bin and every positive regularization there is exactly one best object, given by a closed formula.
2. The error of that estimator is an *identity*: the difference between the estimate and the true bin is exactly a projected leftover minus a bias term, over a positive denominator.
3. The method has *physical blind spots* that no amount of regularization removes—modes outside the aperture, a global phase, a single image of a general complex object, and the interior zeros of a single-defocus contrast-transfer function.

Those are the claims Lean locks down. A later implementation can be checked against those certificates, not only against a plot. The numerical method remains the method; Lean is the certificate that we are running the method we think we are running.

8 What Lean certifies, so the numerical method is trustworthy

The discrete linear stage is recommended because the following statements have been checked, not because a reconstruction on one stack looked clean. They are listed in the order a microscopist meets them. Full statements are in [Appendix A](#).

1. **Same forward model.** The reconstruction uses the bilinear kernel of Yu *et al.* and the identity [\(4\)](#): R_{FO} and R_{CTF} coincide on the slice $\mathbf{q}' = \mathbf{0}$. Stage 1 inverts that slice, not a generic blur and not a substitute propagator.
2. **A well-posed discrete problem.** After a discrete Fourier transform, each spatial frequency decouples. For a positive regularization parameter the energy that measures “predicted minus measured, plus a small penalty on the object” has a unique global minimizer, given in closed form. The computer is not choosing among several equally good objects.
3. **An exact error formula.** If the data equal the linear model plus a leftover n (camera noise, discretization, and the part of FO that the linearization dropped, lumped together), the estimation error is exactly (projected leftover minus a regularization bias) over a positive denominator. This is algebra, not a simulation, and it is not a statistical noise theorem: n need not be random or Gaussian.
4. **A usable bound.** The size of that error is no worse than a simple triangle inequality, and the bound is attained on an explicit example. That is the certificate that regularization plus stacked defoci controls the mistake.
5. **Physical limits, not software bugs.** Frequencies outside the contrast aperture have a vanishing slice at every defocus, so those bins reconstruct as zero. Intensity is unchanged if the exit wave is multiplied by a global phase. One image never identifies a general complex

pair $(X(\mathbf{q}), X(-\mathbf{q}))$. The weak-phase contrast-transfer factor $\sin(2\pi\chi_S)$ vanishes on rings inside a large enough aperture of a *pure-defocus* column ($C_3 = C_5 = 0$, $\Delta z \neq 0$); those holes are why one stacks at least two independent defoci. (That is not the same as a column with spherical aberration still present.) A failed reconstruction at those places is predicted.

6. **When the linear method is allowed.** The FO image of a small change around a flat mirror is exactly “vacuum intensity, plus a term linear in the change, plus a quadratic remainder”. Stage 1 inverts the linear term. If the remainder is large (strong phase), the linear inverse is biased—that is the license for stage 2. If the remainder sits at the noise floor, stage 2 cannot help.
7. **A cost model, not a benchmark.** Counting a modelled discrete Fourier transform as $N \log_2 N$ operations, the linear stage is the cheap method that still uses the whole stack and still fills contrast-transfer zeros. Already on a 128-pixel grid it is strictly cheaper than summing every frequency pair. Lean does *not* prove that a fast Fourier transform exists, or that a laptop will take twenty milliseconds.
8. **Explicitly not proved.** Alignment of the experimental stack; the map from lens current to Δz ; equality of the array transform with the continuum Fourier transform of the paper; Gaussian or Poisson statistics and Cramér–Rao bounds; convergence of the nonlinear refinement; and any claim to be fastest among all conceivable algorithms.

The certificates live in `LeemFO/Tikhonov.lean` and `LeemFO/LinearInverse.lean`; the sinusoid mode count and Bessel expansion live in `LeemFO/Inverse.lean` and `LeemFO/PhaseObject.lean`. [Appendix A](#) names the individual theorems.

9 Five ways one might invert, in plain language

Write N for the number of image pixels and K for the number of focus settings. A typical LEEM stack is $N = 512^2$ and $K \sim 20$.

The words *regularization* and *Tikhonov* will appear once, then we stop leaning on them. Inverting a filter that can be nearly zero is unstable: a small error in a dark frequency becomes a huge error in the object. Tikhonov’s method [17]—the same idea as Wiener’s filter in signal processing [18]—adds a small positive floor α in the denominator so one never divides by zero. The price is a systematic bias of size α , which [Section 8](#) already accounts for.

1. Single-image Wiener / contrast-transfer deconvolution

Take one picture, transform it to frequencies, divide by the filter, and transform back. This is the fastest method, and it is the wrong inverse. The filter zeros punch holes in the spectrum, and a general complex object is not identified by one plane ([Section 4](#)).

2. Transport of intensity

Teague [21] and Paganin and Nugent [22] recover phase from two or three closely spaced defoci by solving a Poisson equation, under the assumption that intensity and phase obey a simple paraxial continuity equation. After 15 kV acceleration a LEEM has $\lambda \sim 10$ pm, a hard aperture, spherical aberration, and partial coherence. Transport of intensity ignores all of that. It produces a low-pass phase map, not an inverse of (3), and it has not been the method used on experimental LEEM graphene or steps.

3. Multi-defocus regularized inversion (Schiske)

Transform every plane to frequencies. Each spatial frequency $\mathbf{q}$ can then be treated on its own. The K complex numbers $\hat{I}_k(\mathbf{q})$ are K noisy readings, each multiplied by the known slice

$$h_k(\mathbf{q}) = R_{\text{FO}}(\mathbf{q}, 0, \Delta z_k). \quad (5)$$

If one pretends the unknown is a single complex amplitude $X(\mathbf{q})$, the regularized estimate is the weighted average

$$\hat{x} = \frac{\sum_k \overline{h_k} y_k}{\alpha + \sum_k |h_k|^2}. \quad (6)$$

The honest linearization around a flat mirror couples $X(\mathbf{q})$ with its partner $X(-\mathbf{q})$ (Section 4); that is a 2×2 regularized solve with the same floor α , unique for every $\alpha > 0$. Equation (6) is the one-unknown special case. If one focus has $h_k = 0$, the others can still determine the bin. That is what “filling contrast-transfer zeros by stacking defoci” means. The cost is one transform per plane, a few arithmetic operations per frequency, and one inverse transform: milliseconds to a few tens of milliseconds on a laptop for $N = 512^2$ and $K \sim 20$. This is Duden’s algorithm [12] with the Yu 2019 envelopes, and it is the unique minimizer Lean already certified (Section 8). It is also exactly one step of the nonlinear method below, started from a flat mirror.

4. Gerchberg–Saxton / hybrid input–output

Gerchberg and Saxton [19] and Fienup [20] iterate between planes, replacing the computed modulus by the measured one and propagating with a coherent lens model. Under partial coherence the correct intensity constraint is the quadratic set of (3), not a coherent modulus. Naïve modulus replacement is therefore the coherent CTF model, and it is biased when FO and CTF disagree off axis [10]. If the iteration is rewritten to project onto the FO intensity, it becomes a slower, undamped version of method 5.

5. Nonlinear least squares on bilinear FO

Guess an exit wave, predict the whole stack with (3), and adjust the guess until the prediction matches the measurements. Each adjustment (a Gauss–Newton step) uses the pair kernel, not only the slice (5), so it sees the mixing that the linear method drops. That is the inverse of the kernel Yu *et al.* actually used, and it is the right estimator for strong-phase objects. It is no longer a per-frequency weighted average about a generic background, so it is slower: seconds rather than milliseconds at typical aperture sizes, for a handful of iterations started from method 3. If the aberrations themselves are unknown, the same idea stacked over extra defoci is the maximum-likelihood (MAL) reconstruction of Coene, Thust, Op de Beeck and Van Dyck [16]. MAL is needed only when the lens coefficients are part of the unknown.

10 Verdict

Method	Typical cost	Fills filter zeros	Faithful for large ϕ
1. Single-image Wiener	one transform	no	no
2. Transport of intensity	one transform	n/a (wrong model)	no
3. Multi-defocus regularized	$K+1$ trans-forms	yes ($K \geq 2$)	linearized only
4. Gerchberg–Saxton / HIO	many forms	trans- if $K \geq 2$	only if the projector is FO
5. Nonlinear least squares on FO	seconds (pair sums)	yes	yes
Jacobi–Anger fit (1D sinusoid)	tiny linear algebra	n/a (parametric)	yes

A modelled operation count for method 3, compared with a dense pair-by-pair apply, is stated in [Appendix A](#).

Fastest usable method for a general 2D experimental stack. Method 3: multi-defocus regularized inversion on the contrast-transfer slice. It is the cheapest algorithm that (i) reads all KN frequency samples, (ii) fills interior filter zeros by stacking defoci, and (iii) has a unique minimizer for every $\alpha > 0$.

Best FO-faithful method. Method 3 as the starting guess, then a few damped Gauss–Newton steps on method 5 (or MAL [16] if aberrations are unknown). Skip the nonlinear stage when the FO residual of method 3 is already at the noise floor—equivalently $\max|\phi| \lesssim 0.3$ at typical LEEM signal-to-noise, i.e. topography $A \lesssim 0.01$ nm at $E_0 = 10$ eV.

Fastest still-correct method for Yu’s sinusoid. Fit the phase amplitude φ on the $M = 2\lfloor q_{\text{ap}}\Lambda \rfloor + 1$ Bessel modes that the aperture keeps ([Section 13](#)). That is method 5 restricted to a one-parameter family.

11 Stage 1: the discrete recipe

The following steps are the unique minimizer of [Section 8](#), written as an experimental procedure.

1. Align the stack $I_k(\mathbf{r})$ by rigid shifts. Alignment is preprocessing; it is not part of the certificate.
2. Subtract the spatial mean of each plane and take a discrete Fourier transform. The zero-frequency bin carries total flux ($R_{\text{FO}}(0, 0) = 1$) and is handled separately.
3. For every frequency $\mathbf{q}$ with $|\mathbf{q}| \leq q_{\text{ap}}$ and every recorded defocus k , evaluate the closed-form slice $h_k(\mathbf{q}) = R_{\text{FO}}(\mathbf{q}, 0, \Delta z_k)$ from the Yu 2019 envelopes, using column λ , χ in waves, and the $+\frac{1}{2}\Delta z\lambda q^2$ sign.
4. At that bin, solve either the one-unknown average (6) or the conjugate-pair 2×2 regularized system (same $\alpha > 0$; the 2×2 system is not (6)).
5. Set the reconstruction to zero for $|\mathbf{q}| > q_{\text{ap}}$.
6. Inverse-transform the assembled spectrum and read $\sigma = |\hat{\psi}|$ and $\phi = \arg(\hat{\psi})$ in the vacuum gauge: the zero-frequency component real and positive.

Units. The kernel R_{FO} uses column wavelength λ (after acceleration). Object phase uses λ_0 : $\phi = 4\pi A/\lambda_0$ for a topographic height A in reflection.

Defocus sign. This library uses $+\frac{1}{2}\Delta z\lambda q^2$ in the wave aberration. Experimental lens current must be mapped onto that convention by one fitted affine transform. Mixing the minus-sign convention of some published simulators reverses overfocus and underfocus.

12 When stage 1 is biased

Kinematic LEEM phase on reflection is (2). At $E_0 = 10\text{ eV}$, $A = 0.1\text{ nm}$ already gives $\phi \sim 3$. The linear inverse never sees off-axis pairs $(\mathbf{q}, \mathbf{q}')$ where FO and CTF disagree. That dropped piece is exactly the quadratic remainder Lean certified in Section 8: the FO image of the increment itself. For a strong-phase sinusoid it is the whole image [10].

Skip test. Predict the stack from the stage-1 reconstruction using the full FO kernel (3). If that residual is consistent with the noise, the quadratic products are undetectable and stage 2 cannot help. Do not skip merely because the slice (4) looks faithful: stage 1 never saw the off-axis pair envelopes $R_{\text{FO}}(\mathbf{q}, \mathbf{q}')$ with both frequencies nonzero.

13 Stage 2, and the one-dimensional sinusoid

Stage 2 is the sentence “adjust the guess until the predicted images match the stack,” with the prediction computed from the full pair kernel (3). Started from stage 1, a few damped Gauss–Newton steps are the usual implementation. Convergence of that iteration is informal; the license to run it is the quadratic remainder of Section 8.

If the object is known to be a single sinusoid of period Λ (Yu’s example), the unknown collapses to an amplitude φ and a shift x_0 . The period is read from the fundamental of the intensity. The aperture keeps only

$$M = 2n_{\text{ap}} + 1, \quad n_{\text{ap}} = \lfloor q_{\text{ap}}\Lambda \rfloor \quad (7)$$

modes. The FO image is then a double sum of M^2 terms, by the Jacobi–Anger expansion

$$e^{i\varphi \sin(2\pi x/\Lambda)} = \sum_{n \in \mathbb{Z}} J_n(\varphi) e^{2\pi i n x/\Lambda}. \quad (8)$$

Fitting φ with the analytic derivative $J'_n = \frac{1}{2}(J_{n-1} - J_{n+1})$ is a tiny least-squares problem. Lean proves the mode count, the expansion, and the decay of the dropped Bessel amplitudes. This reduction does *not* apply to a general 2D micrograph: there is no Bessel ladder, and stage 2 stays a Gauss–Newton iteration in the aperture basis.

14 Conclusion

For a 2D experimental LEEM focal series, recover the object in two stages. Stage 1 is a regularized, frequency-by-frequency inversion of the through-focal stack on the contrast-transfer slice of the Yu 2019 kernel. It fills the holes a single focus leaves, it has a unique answer for every positive regularization, and it is the cheapest method that still does those two things. Stage 2 is a nonlinear refinement on the full Fourier-optics kernel, used only when the stage-1 prediction disagrees with the stack by more than the noise—the strong-phase regime of ripples and steps. For the one-dimensional sinusoid that exhibits contrast-transfer failure, stage 2 collapses to a Bessel fit.

The linear stage is recommended because, under the Yu 2019 forward model, it is well-posed: unique, equipped with an exact error identity and a sharp bound, honest about aperture, gauge, single-image, and filter-zero blindness, and first-order consistent with bilinear Fourier optics. Lean is the certificate of those statements, not a reason the method happened to converge on one dataset. Yu *et al.* forward-match; Duden *et al.* invert linearly; the two-stage pipeline does both.

A Formal claims

This appendix is optional. The statements are finite-dimensional: a finite list of defoci and a finite grid of frequencies. Noise n_k is a deterministic leftover (linearization remainder, instrument noise, and discretization), not a random variable. No existence theorem for the fast Fourier transform is claimed.

Theorem A.1 (Unique minimizer). *For $\alpha > 0$ the energy $J(x) = \sum_k \|h_k x - y_k\|^2 + \alpha \|x\|^2$ has a unique global minimizer (6). Completing the square: $J(x) = J(\hat{x}) + D \|x - \hat{x}\|^2$ with $D = \alpha + \sum |h_k|^2 > 0$.*

Lean: `tikhonovJ_min`, `tikhonov_unique`, `tikhonovJ_eq_shift`. The same identity gives uniqueness at $\alpha = 0$ whenever $\sum |h_k|^2 > 0$.

Theorem A.2 (Bias–noise identity). *If $y_k = h_k x^* + n_k$ and $D \neq 0$, then*

$$\hat{x} - x^* = \frac{\sum_k \overline{h_k} n_k - \alpha x^*}{D}. \quad (9)$$

Lean: `tikhonov_error`. Taking $\mathbb{E}[n] = 0$ is informal (no probability space).

Theorem A.3 (Triangle bound, sharp). *If $\alpha \geq 0$ and $D > 0$,*

$$|\hat{x} - x^*| \leq \frac{\sum_k |h_k| |n_k| + \alpha |x^*|}{D}. \quad (10)$$

Equality holds for one measurement, $h = n = 1$, $x^ = -1$, $\alpha = 1$.*

Lean: `tikhonov_error_bound`, `tikhonov_error_bound_sharp`.

Theorem A.4 (Quadratic remainder of bilinear FO). *With vacuum spectrum $1_{\mathbf{q}=0}$ and increment δ ,*

$$\hat{I}(\text{vac} + \delta) = \hat{I}(\text{vac}) + DI_{\text{vac}}[\delta] + \hat{I}(\delta), \quad (11)$$

where $DI_{\text{vac}}[\delta](\xi) = R_{\text{FO}}(\xi, 0) \delta(\xi) + R_{\text{FO}}(0, -\xi) \overline{\delta(-\xi)}$ and $|\hat{I}(\delta)(\xi)| \leq (\sum |R_{\text{FO}}|)(\sum |\delta|)^2$.

Lean: `ihat_add`, `ihatJac_vacuum`, `ihat_quadratic_remainder`, `ihat_bound`. The dropped term is exactly bilinear: linearized inversion is biased by $O(\|\delta\|^2)$, which is the whole image for a strong-phase sinusoid [10].

Theorem A.5 (Interior CTF zeros of a pure-defocus column). *$\sin(2\pi\chi_S) = 0$ whenever $2\chi_S \in \mathbb{Z}$. For $C_3 = C_5 = 0$, $\Delta z \neq 0$, and a large enough aperture, there exists nonzero q inside the disk with that vanishing (*exists_interior_weakPhase_zero*). A single defocus is therefore blind at those rings; $K \geq 2$ with incommensurate Δz typically fills them. This is a pure-defocus hypothesis, not a column that still has spherical aberration.*

The 2×2 Gram determinant of the vacuum Jacobian is positive for $\alpha > 0$ (`gramDet_pos`), so the conjugate-pair normal equations are invertible even though one measurement is not (Section 4).

Modelled cost. Set $\text{dftCost}(N) := N \log_2 N$ and

$$\text{reconstructCost}(K, N) = (K + 1) N \log_2 N + N(8K + 4). \quad (12)$$

For $K \geq 1$ and $N \geq 2$, $\text{reconstructCost}(K, N) \leq 14 K N \log_2 N$. At $N = 128$ this is strictly cheaper than a dense bilinear apply KN^2 , for every $K \geq 1$ (`reconstructCost_le`, `reconstructCost_lt_dense_128`). The constant 14 is attained in the model at $K = 1$, $N = 2$.

Claim	Lean
Tikhonov unique min, closed form	<code>LeemF0/Tikhonov.lean</code>
Bias–noise identity and sharp triangle bound	same
Cost model versus dense KN^2	<code>reconstructCost_*</code>
Aperture support of $R_{\text{FO}}(\cdot, 0)$	<code>LeemF0/LinearInverse.lean</code>
Gauge, vacuum Jacobian, quadratic remainder	same
$K = 1$ not injective; pure-defocus CTF zeros	same
Mode count $M = 2\lfloor q_{\text{ap}}\Lambda \rfloor + 1$	<code>LeemF0/Inverse.lean</code>
Jacobi–Anger and Bessel decay	<code>LeemF0/PhaseObject.lean</code>
FFT exists / equals the continuum FT	informal
Gaussian/Poisson noise, Cramér–Rao	informal
Gauss–Newton local quadratic convergence	informal
Image alignment, current-to-defocus map	informal
“Fastest among all conceivable algorithms”	only DFT versus dense bilinear

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